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SOURCING & PROVENANCE DOSSIER

Sourcing Pure Agarwood Oil

Why cultivated Malaysian Aquilaria malaccensis is the responsible perfumer's standard

PREPARED FOR

Luxury Perfumers · Boutique Fragrance Houses · High-End Wellness Brands

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1. The "Liquid Gold" Market

Agarwood — *oud*, *gaharu*, *aloeswood* — is the dark, resin-saturated heartwood that forms when a tree of the genus *Aquilaria* defends itself against wounding and infection. The healthy tree is pale and unremarkable; only when injury allows fungi and pathogens to enter does it flood the affected tissue with aromatic resin that hardens, over years, into one of the most valuable raw materials on earth. Distilled, that resin yields oud oil — a material that has commanded comparison to gold by weight, with fine grades reaching tens of thousands of US dollars per litre.

Its cultural reach is unusually deep. Oud anchors the great attars of the Gulf, perfumes the courts and mosques of the Islamic world, scents Japanese kōdō incense ceremony, and now sits at the heart of a Western fine-fragrance category that did not meaningfully exist twenty years ago. For a house building around oud, the material is not an ingredient among many — it is the signature, and its provenance is the story.

> **The scarcity at the centre of the value.** Only a small fraction of wild Aquilaria trees — on the order of one in ten — ever form agarwood naturally. That single fact explains both the price and the problem: genuine wild oud is rare enough that the market around it has filled with substitution, adulteration and outright fakery.

2. The Problem with Wild Harvest

For a luxury house, wild-sourced oud is no longer a mark of authenticity — it is a liability on three fronts: legal, ethical and qualitative.

2.1 The CITES position

Aquilaria malaccensis — the principal and most prized source species, and the one native to Malaysia — was listed on **CITES Appendix II in 1995**; in 2004–2005 the listing was extended to cover all *Aquilaria* species. Appendix II means international trade is permitted only under a system of export permits and non-detriment findings designed to ensure trade does not threaten the species' survival. The species is simultaneously classified as **Vulnerable** on the IUCN Red List. In practical terms: any oud crossing a border without correct CITES documentation is illegal, and a fragrance house handling it inherits that exposure.

2.2 Environmental degradation

Because resin-bearing trees cannot be identified from the outside, wild harvesting is destructive by nature: trees are felled speculatively in the hope of finding resin, and historically the great majority were cut in vain. Across the species' range, decades of this have stripped once-abundant stands and driven the population declines that prompted the CITES and IUCN listings in the first place. Illegal harvesting and smuggling persist despite the controls, and they are precisely what a credible luxury brand cannot afford to be connected to.

2.3 The quality problem

Wild material is also inconsistent by definition. Resin content, age, the part of the tree, the infecting organism and the harvest method all vary uncontrollably, so wild oud arrives as a lottery of grades. That inconsistency is the gap into which adulteration flows — and for a perfumer who must reproduce an accord batch after batch, an unrepeatable raw material is a commercial hazard regardless of how beautiful any single lot may be.

3. The Plantation Advantage

The response to scarcity has been cultivation, and Malaysia is one of its centres. Across South, East and Southeast Asia, large *Aquilaria* plantations now exist; since 2017, CITES trade data has recorded **more plantation-origin than wild-origin** agarwood in international trade — a structural shift in where the world's oud comes from.

3.1 Controlled inoculation: turning a lottery into a process

In the wild, agarwood forms when ants and wood-boring insects introduce fungi through natural wounds and the tree responds by laying down aromatic resin. Plantation cultivation reproduces this deliberately: mature trees are wounded under controlled methods — drilling and the introduction of identified fungal inoculants — to trigger the same defensive resin formation, on a known timetable, in trees of known age and provenance. Decades of trial and refinement have produced repeatable best-practice protocols.

The consequences for a buyer are direct:

- **Consistency.** Known species, age, inoculation method and harvest window produce resin — and therefore oil — of far more uniform character batch to batch.
- **Traceability.** A specific plantation, even a specific block of trees, can stand behind a given lot, with the CITES documentation that legal cultivated trade requires.
- **Ethical security.** No felling of wild trees, no speculative destruction, no poaching exposure — and active relief of pressure on remaining wild populations.
- **Supply continuity.** A planted, managed resource can be scheduled and scaled in a way a dwindling wild one cannot.

> **Sustainability is now the premium, not the compromise.** The old assumption that wild oud is inherently finer no longer holds. Modern cultivated Aquilaria malaccensis, correctly inoculated and matured, delivers the depth perfumers want with the consistency and clean provenance that wild material cannot — and that a discerning clientele increasingly demands.

4. Scent Profile: Malaysian Aquilaria malaccensis

The character of oud lives in its chemistry. Agarwood oil is dominated by **sesquiterpenes** and aromatic compounds — frequently together accounting for the great majority of the oil — and it is these molecules that build the scent. The woody, balsamic core comes from sesquiterpene hydrocarbons such as α-gurjunene and α-guaiene; α-copaene lends a spicy-woody facet; and a family of oxygenated sesquiterpenes (agarospirol, jinkohol, the eudesmols) and 2-(2-phenylethyl)chromones supply the deep, resinous, almost animalic warmth that distinguishes true oud.

Origin shapes the accord. Analytical work across Southeast Asian agarwoods shows that **Malaysian material tends to be rich in alcohols and aldehydes**, a profile associated with high-quality, refined aromatic character — distinct from the heavier sesquiterpene dominance of some other origins. A useful authentication marker also points home: *10-epi-γ-eudesmol*, a major sesquiterpenoid in A. malaccensis oil, helps validate the species of origin.

The olfactory signature

In the language of the perfumer's organ, fine Malaysian malaccensis reads as a deep, complex woodiness wrapped in balsamic sweetness, with a warm, faintly animalic undertone and a resinous, long-lasting drydown. It is this bittersweet, enveloping warmth — woody, balsamic and faintly leathery at once — that separates genuine oud from synthetic reconstructions, which tend to read as flatly woody and leathery without the living, balsamic aura beneath.

5. Verifying Purity: Buying with Confidence

The oud market's greatest hazard is not price — it is authenticity. Dilution with carrier oils, blending with cheaper Aquilaria species, and outright synthetic "oud" bases are widespread. The good news is that authenticity is now verifiable, and a serious buyer should insist on the evidence.

5.1 The analytical baseline: GC-MS

Gas chromatography–mass spectrometry (**GC-MS**) is the definitive tool. It separates and identifies the oil's constituent compounds, allowing a laboratory to confirm the sesquiterpene and chromone signature of genuine agarwood, to flag species via marker compounds such as *10-epi-γ-eudesmol* for malaccensis, and to expose adulterants — carrier oils, foreign terpenes, or synthetic molecules that have no place in a pure distillate. A reputable supplier should be able to provide a GC-MS profile for the lot you are buying.

5.2 A practical verification checklist

1. **Demand a GC-MS report** for the specific batch, showing the characteristic sesquiterpene/chromone profile and species marker.
2. **Require CITES documentation** evidencing legal, plantation origin — essential for Appendix-II material and for your own import compliance.
3. **Confirm species and origin in writing** — Aquilaria malaccensis, named plantation, Malaysia — not merely "oud, Southeast Asia."
4. **Assess organoleptically** for the living balsamic warmth and evolving drydown that synthetics cannot reproduce.
5. **Verify it is neat oil** — a pure, undiluted distillate — and that any dilution, if offered, is disclosed and quantified.

> **The R&J position.** We supply oud from cultivated Malaysian Aquilaria malaccensis: a single, named, prized species; controlled inoculation for a consistent and characteristic profile; full CITES documentation for legal, plantation-origin trade; and analytical backing on request. For a house whose name will sit beside the word "oud," that is the difference between a beautiful story you can defend and a risk you cannot.

Disclaimer. *This dossier is provided for commercial and educational information and does not constitute legal or regulatory advice. International trade in Aquilaria species is governed by CITES (Appendix II) and by national implementing laws that vary by jurisdiction and change over time; buyers are responsible for verifying current import requirements and securing valid permits. Olfactory descriptions are interpretive. Confirm species, origin and purity through documentation and independent analysis.*

Key sources. *CITES (Appendix II listings, 1995 / 2004–2005; CoP19 documentation on Aquilaria and Gyrinops); IUCN Red List (Aquilaria malaccensis, Vulnerable); TRAFFIC, "Heart of the Matter"; peer-reviewed GC-MS analyses of A. malaccensis essential oil (sesquiterpene and chromone profiling; SPME/GC-MS olfactometry studies).*